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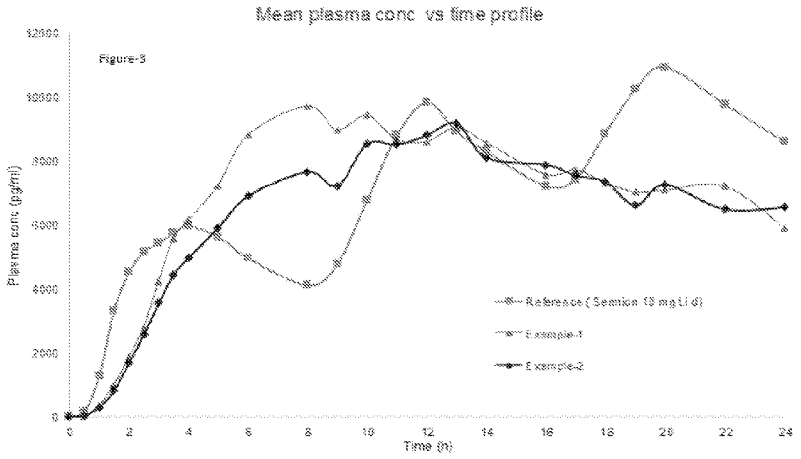
Figure-3



(57) Abstract: The present application relates to a modified release composition comprising nicergoline, or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients. The present application also relates to a modified release composition that provides steady plasma concentrations with minimal fluctuations and offer convenient dosing (i.e., dosed less frequently).

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MODIFIED RELEASE NICERGOLINE COMPOSITIONS

FIELD OF THE APPLICATION

[1] The present application relates to a modified release composition comprising nicergoline or its pharmaceutically acceptable salt thereof.

BACKGROUND OF THE APPLICATION

[2] Nicergoline is an ergot derivative used to treat senile dementia and other disorders with vascular origins. The product is indicated to prevent functional declines due to the sequelae of cerebral infarction and for the treatment of cognitive impairment in various forms of dementia.

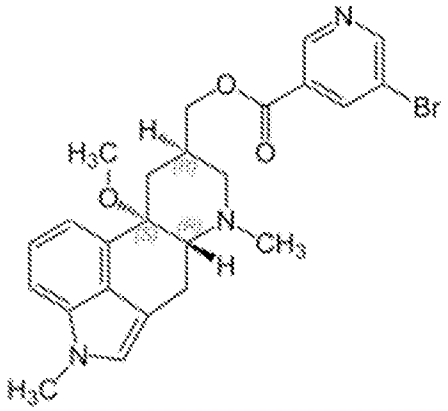
[3] Nicergoline and its active metabolites favorably influence the cerebral hemodynamic and metabolism. These activities are due to the α-lytic and myolytic activities and to the increase in glucose and oxygen consumption. Nicergoline undergoes rapid metabolism and is undetectable in serum by conventional bioassay methods. However the drug's two major metabolites 1-methyl-10-methoxy-dihydro-lysergol (MMDL) and 10-methoxy-dihydro-lysergol (MDL) are quite stable and are detectable.

[4] Nicergoline is absorbed in the intestinal tract and rapidly metabolized by hydrolysis, followed by partial demethylation and/or conjugation with glucuronic acid.

[5] In humans, the administered dose is excreted in metabolized form, mainly in the urine and in a minor amount (10%) in the feces in a period of 3-4 days. Nicergoline is used for the clinical treatment of acute and chronic cerebro-vascular metabolic disorders due to arteriosclerosis thrombosis, cerebral embolism and transient cerebral ischemia.

[6] Nicergoline is currently marketed as Sermion® by Pfizer in most of the countries as tablet (5, 10, 15 & 30 mg), capsule (5 & 10 mg), injectable (4, 5 and 8 mg) liquid/drops (10 mg & 5 mg/ml), powder, and suspension (300 mg) dosage forms.

[7] The CAS Registry Number for nicergoline is 27848-84-6 and the chemical name is [(8β)-10-methoxy-1,6-dimethylergolin-8-yl]methyl 5-bromopyridine-3-carboxylate; while its structural formula is:



[8] Its empirical formula is $C_{24}H_{26}BrN_3O_3$, and its molecular weight is 484.386.

[9] Nicergoline is generally administered orally 2-3 times daily, for several months. This normally leads to adverse drug reactions such as hypotension, dizziness, stomach pain, thereby decreasing patient compliance.

[10] Once-a-day modified release nicergoline composition would bring about remarkable advantages in terms of convenience for the patient and can improve the compliance. Moreover, it is known that from the therapeutic point of view a controlled-release composition allows to maintain more constant and lower blood concentrations, thus decreasing possible side-effects of nicergoline such as low blood pressure and dizziness, stomach disorders, hot flashes and skin redness. Modified-release compositions are particular pharmaceutical compositions which have the ability to maintain a desired blood level of a medicament over an extended period of time by minimizing the peak-trough variations in plasma concentrations. It is therefore desirable to administer nicergoline through modified-release compositions, having improved dissolution properties and extended release profiles.

[11] Several attempts were made to prepare modified release nicergoline compositions, which include EP0602619 A1, CN 101269035 and CN 101095663.

[12] However, since aqueous solubility of nicergoline is very low, it is difficult to prepare as modified-release compositions. It is often difficult to predict whether a particular modified-release composition will provide the desired release profile for a relatively insoluble compound.

[13] Accordingly, there is a long felt need for a modified release nicergoline composition which provides steady plasma concentrations with minimal fluctuations and to provide more convenient dosing (i.e., dosed less frequently). Further it is desired to have a composition that produces less peak to trough plasma concentration fluctuation over the dosing interval, while providing comparable total exposure and less or comparable peak exposure to the active drug ingredients (and metabolites).

SUMMARY OF THE APPLICATION

[14] Thus, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, which is convenient in dosing and offers less peak to trough plasma concentration fluctuation over the dosing interval.

[15] The present application also relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients.

[16] In an embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other

pharmaceutically acceptable excipients, wherein said composition provides therapeutically effective amount of nicergoline for a period of about 24 hours after oral administration.

[17] In an aspect of the above embodiment, the rate-controlling polymer comprises at least one hydrophilic polymer in an amount of from about 10% to about 70% by weight based on the total weight of the composition.

[18] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising about 30 mg to about 60 mg nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said rate-controlling polymer comprises at least one hydrophilic polymer in an amount of from about 10% to about 70% by weight based on the total weight of the composition, wherein said composition provides therapeutically effective amount of nicergoline for a period of about 24 hours after oral administration.

[19] In an aspect of the above embodiment, the composition of the present application may be in a monolithic matrix form and comprises intra-granular and extra-granular portions of components.

[20] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said composition is in a monolithic matrix form and comprises intra-granular and extra-granular portions of components.

[21] In one aspect of the above embodiments, the composition of the present application is in a monolithic matrix form comprising at least one hydrophilic polymer present in both extra-granular portion and intra-granular portions, in a weight ratio of about 0:1 to 1:1.

[22] In one embodiment, the composition of the present application exhibits at least one of the following dissolution profiles: (i) releases no more than about 30% of nicergoline in 2 hours, in 900ml of 0.1 N HCl solution, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C, (ii) releases no more than about 30% of nicergoline in 2 hours, in 900ml of acetate buffer having pH of 4.5, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C, (iii) releases no more than about 30% of nicergoline in 2 hours, in 900ml of phosphate buffer having pH of 6.8, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C and (iv) releases no more than about 30% in 2 hours, no more than about 60% in 6 hours, and more than about 70% in 10 hours of nicergoline when subjected to a changeover dissolution study.

[23] In another embodiment, the composition of the present application when administered as single dose exhibits bioequivalence to a commercially available 10 mg nicergoline immediate release composition when administered thrice daily to a human subject in fasting conditions and said bioequivalence is established by: (i) a 90% Confidence Interval for mean $AUC_{(0-t)}$, which is between 80% and 125%, and (ii) a 90% Confidence Interval for mean $AUC_{(0-\infty)}$, which is between 80% and 125%.

[24] In an aspect of the above embodiment, the composition of the present application when administered as a single dose to a human subject in the fasting and fed conditions have a mean AUC_{0-4} in the ratio of about 1:1.3.

[25] In another aspect of the above embodiments, the composition of the present application when administered as a single dose to a human subject in the fasting conditions have an exposure up to 8, 16 and 24 hours similar to an immediate release formulation comprising nicergoline 10 mg administered thrice daily in fasting conditions and have the mean AUC_{0-8}, AUC_{0-16} and AUC_{0-24} in the ratio of about 1:1.

[26] In another aspect of the above embodiments, the composition of the present application when administered as a single dose to a human subject in the fasting conditions have a mean AUC_{0-8} to AUC_{0-2} in the ratio of about 1: 0.03.

[27] In another aspect of the above embodiments, the composition of the present application when administered as a single dose to a human subject in the fasting conditions have an average peak to trough plasma concentration ratio of simulated steady state profiles of about 1.66:1.

[28] In another aspect of the above embodiments, the composition of the present application when administered as a single dose to a human subject in the fasting conditions and in the fed conditions have a plasma concentration profile at 2 hours in the ratio of about 1:1.

[29] In one embodiment, the composition of the present application following a single dose oral administration to a human subject in fasting conditions exhibit at least one of the following plasma profiles: (i) about 6% of nicergoline absorption profile at 2 hours, (ii) about 20% of nicergoline absorption profile at 4 hours, (iii) about 42% of the nicergoline absorption profile at 8 hours and (iv) about 58% of the nicergoline absorption profile at 12 hours.

[30] In another embodiment, the composition of the present application has delayed Tmax in both fasting and fed conditions when compared with first dose of the three divided doses of commercially available 10 mg nicergoline immediate release composition.

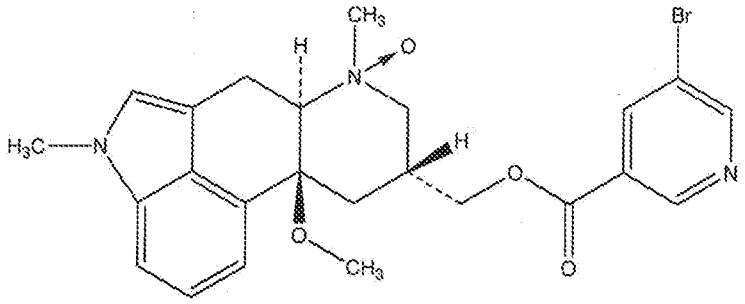
[31] In an aspect of the above embodiment, the composition has Tmax in the range of about 5-20 hours or about 8-15 hours or about 10-15 hours or about 13 hours.

[32] In one embodiment, the composition of the present application when administered to a human subject under fasting conditions provides steady state plasma profile and exhibits at least one of the following plasma profiles: (i) C_{max} of about 5 ng/ml to 19 ng/ml of nicergoline or its detectable metabolites, (ii) $C_{min,ss}$ is about 8 ng/ml to about 15 ng/ml of

nicergoline or its detectable metabolites, (iii) $C_{max,ss}$ is about 13 ng/ml to about 26 ng/ml of nicergoline or its detectable metabolites, and (iv) $C_{avg,ss}$ is about 10 ng/ml to about 18 ng/ml of nicergoline or its detectable metabolites.

[33] In another embodiment, the composition of the present application remains physically and chemically stable, upon storage at a temperature of about 40°C and relative humidity of about 75% for at least three months, and produces not more than 3% of total relative substances.

[34] In an aspect of the above embodiment, the composition upon storage at a temperature of about 40°C and relative humidity of about 75% for three months, remains physically and chemically stable, produces not more than 3% of total relative substances, and produces less than 0.5% w/w of nicergoline N-Oxide compound having the structure of:



[35] In an aspect of the above embodiments, the composition of present application is administered for the treatment of patients having acute or chronic cerebral metabolic-vascular disorders having cognitive loss and/ or dementia.

BRIEF DESCRIPTION OF THE DRAWINGS

[36] Figure. 1 shows dissolution profile of the nicergoline in Example-1 and Example-2.

[37] Figure. 2 shows the plasma concentration of nicergoline metabolite MDL in pg/ml of Example-1 and Example-2 over a period of 24 hours for oral administration of composition containing 30 mg of nicergoline of the present application.

[38] Figure. 3 shows the plasma concentration of nicergoline metabolite MDL in pg/ml of Example-1 and Example-2 over a period of 24 hours for oral administration of composition containing 30 mg of nicergoline of present application given once under fasting condition vs. plasma concentration of nicergoline metabolite MDL for oral administration of reference product (Sermion®) tablets containing 10 mg of immediate release nicergoline (reference product) given t.i.d. under fasting condition.

[39] Figure. 4 shows the plasma concentration of nicergoline metabolite MDL in pg/ml of Example-1 and Example-2 over a period of 72 hours for oral administration of composition containing 30 mg of nicergoline of the present application given once under fasting condition vs. plasma concentration of nicergoline metabolite MDL for oral administration of reference product (Sermion®) tablets containing 10 mg of immediate release nicergoline given t.i.d. under fasting condition.

DETAILED DESCRIPTION OF THE APPLICATION

[40] The details of one or more embodiments of the present invention are set forth in this document. Modifications to embodiments described in this document, and other embodiments, will be evident to those of ordinary skill in the art after a study of the information provided in this document. The information provided in this document, and particularly the specific details of the described exemplary embodiments, is provided primarily for clearness of understanding and no unnecessary limitations are to be understood therefrom. In case of conflict, the specification of this document, including definitions, will control.

[41] Definitions: The terms as used herein have the following meanings:

[42] The terms "a" and "the" as used herein are understood to encompass the plural as well as the singular or otherwise clearly mentioned wherever needed. For example, reference to "an excipient" includes reference to one or more of such excipients, and reference to "the carrier" includes reference to one or more of such carriers.

[43] The term "comprising" (and its grammatical variations) as used herein is meant to be open ended and used in the inclusive sense of "having" or "including" and not in the exclusive sense of "consisting only of." The term "consisting essentially of", as used herein means the claimed elements and others, but is meant to exclude things that are inconsistent with the basic and novel characteristics of the inventions.

[44] The terms such as 'about', 'up to', 'generally', 'substantially' and the like are to be construed as modifying a term or value such that it is not an absolute. Such terms will be defined by the circumstances and the terms that they modify as those terms are understood by those of skill in the art. This includes, at very least, the degree of expected experimental error, technical error and instrumental error for a given experiment, technique or an instrument used to measure a value. The term "about" is used to provide flexibility to a numerical range endpoint by providing that a given value may be "a little above" or "a little below" the endpoint. As used herein, the term "about" means a slight variation of the value specified, within 10 percent of the value specified. Nevertheless, the term "about" can mean a higher tolerance of variation depending on for instance the experimental technique used. Said variations of a specified value are understood by the skilled person and are within the context of the present invention. As an illustration, a numerical range of "about 1 to about 5" should be interpreted to include not only the explicitly recited values of about 1 to about 5, but also include individual values and sub-ranges within the indicated range. Thus, included in this numerical range are individual values such as 2, 3, and 4 and sub-ranges such as from 1-3,

from 2-4, and from 3-5, etc., as well as 1, 2, 3, 4, and 5, individually. This same principle applies to ranges reciting only one numerical value as a minimum or a maximum.

[45] The term "pharmaceutically acceptable" salt includes derivatives of the disclosed compounds which are, within the scope of sound medical judgment, suitable for use in humans and lower animals without undue toxicity, irritation, allergic response and the like, which are well known in the art. The salt can be prepared in situ during the final isolation and purification of the compounds of the invention, or separately by reacting the pharmaceutically active substance, having a freebase function, with a suitable organic acid or inorganic acid. Further refers to pharmaceutically acceptable solvates, including hydrates, of such compounds and such salt.

[46] The term "pharmaceutical composition" or "composition" for purposes of the present disclosure includes solid pharmaceutical products such as tablets, capsules, sachets, pills, or granules, containing a mixture of two or more compounds, elements or molecules. This may be matrix based compositions, reservoir based formulations, multi-particulate based formulations, multi-layer formulations, resin formulations, osmotic formulations, gastro-retentive formulations, etc.

[47] The term “monolithic matrix” or “matrix” as used herein refers to a mixture containing at least one rate-controlling polymer, in which the drug is dispersed or dissolved or intermixed uniformly to have a homogenous or intimate mixture/ dispersion and capable of delivering the drug at a controlled rate for a period of time. The overall release kinetics of the drug from the matrix is generally linear, such that a relatively constant amount of drug is released over the desired time period.

[48] By the term “modified release” (MR) it is meant that an active agent from a dosage form is released in controlled fashion in an environment over (throughout or during) an extended period of time. The term “modified release”, as regards to drug release, includes

the terms like “extended release”, “controlled release”, “prolonged release”, “sustained release”, “slow release”, or alike terms as used in the pharmaceutical sciences.

[49] The term "excipient" or "pharmaceutically acceptable excipient" means a component of a pharmaceutical product that is not having any pharmacological effect, such as a filler, diluent, carrier, etc. The excipients that are useful in preparing a pharmaceutical composition are generally safe, non-toxic and are acceptable for veterinary use as well as human pharmaceutical use. An "excipient" or a “pharmaceutically acceptable excipient" as used in the specification includes both one and more than one such excipients.

[50] The term "free of" or "substantially free of" as used herein refers to the absence of any separately added portion of the referenced compound or composition or excipients.

[51] The term “Cp” as used herein denotes plasma concentrations.

[52] The term “fasting” in t.i.d. dosing (is an abbreviation for "ter in die" which in Latin means three times a day) refers to fasting condition 2 hours before and after the dose administration.

[53] The term “stability” as used herein includes both chemical stability and physical/polymorphic stability. The term ‘stability’ is defined as the capacity of a drug substance or drug product to remain within the established specifications to maintain its identity, strength, quality and purity throughout the retest or expiration dating period. The term ‘chemical stability’ means the tendency of drug to resist change or decomposition due to internal reaction, or due to the effects of oxygen, heat, light, pressure, etc.

[54] The term ‘related substances’ (RS) or ‘impurities’ mean the degradation impurities or active ingredient process related impurities of drug material.

[55] The term “AUC” refers to the area under the plasma concentration-time curve, as calculated by the log-linear trapezoidal rule over the complete dosing interval. The AUC (area under the curve) for a specified period is referred to as a "partial AUC" for the period.

[56] The term “C_{max}” refers to the highest plasma concentration of the drug attained within the dosing interval.

[57] The phrase "$C_{max,SS}$" refers to the highest plasma concentration (e.g., ng/ml) observed after repeated once daily administration at steady state.

[58] The phrase "$C_{min,SS}$" refers to the trough plasma concentration (e.g., ng/ml) observed after repeated once daily administration at steady state.

[59] The phrase "$C_{avg,SS}$" refers to the average plasma concentration at steady state.

[60] The term “T_{max}” refers to the time period which elapses after administration of a single dosage form at which the plasma concentration of the drug attains the highest plasma concentration of drug attained within the dosing interval.

[61] The term ‘shelf life’ is the time that finished products can be stored after manufacturing, during which the defined quality of a specified proportion of the product remains acceptable under expected (or specified) conditions of distribution, storage, and display. The shelf life is established by the manufacturer of a product.

[62] The term “stability testing” refers to tests conducted at specific time intervals and in various environmental conditions (e.g., temperature and humidity) to see if and to what extent a drug product degrades over its designated shelf life time. The specific conditions and time of the tests are accelerated, in order to mimic the conditions the drug product is expected to encounter over its shelf life.

[63] The term “RP-HPLC” refers to Reversed-phase high-performance liquid chromatography.

[64] In an embodiment, the present application relates to a modified release composition of nicergoline or its pharmaceutically acceptable salt thereof.

[65] In one embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said composition provides therapeutically effective amount of nicergoline for a period of about 24 hours after oral administration.

[66] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising from about 30 mg to about 60 mg of nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said composition provides therapeutically effective amount of nicergoline for a period of about 24 hours after oral administration.

[67] In certain embodiments, the application relates to a modified release composition in a monolithic matrix form and delivers nicergoline in a controlled release manner.

[68] In further embodiments, the application relates to a modified release composition for once daily oral administration comprising about 30 mg to about 60 mg of nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said composition is in a monolithic matrix form and provides therapeutically effective amount of nicergoline for a period of about 24 hours after oral administration.

[69] In another embodiment, the drug can be released predominantly by two mechanisms that include diffusion and/or degradation. Diffusion occurs when the drug is released either through pores in the polymer matrix or by passing between polymer chains of the matrix. In a diffusion system, the drug can be dispersed throughout the matrix, or

localized within a reservoir adjacent to or within the matrix. In a reservoir system, a reservoir of drug, for example, solid drug, dilute solution, or highly concentrated drug solution within a polymer matrix is surrounded by a controlled release material through which the drug is able to diffuse. In a degradable system, the drug is released as the matrix is degraded in vivo. Drug can also be released by a combination of the two mechanisms.

[70] In one embodiment, the monolithic matrix described herein releases the drug by a combination of both diffusion and degradation. The release rate can be controlled by varying the percent of rate-controlling polymer or varying the drug to polymer ratio.

[71] In further embodiments, the present application relates to a modified release composition for once daily oral administration comprising about 30 mg to about 60 mg of nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said composition is in a monolithic matrix form and said nicergoline is intermixed uniformly with rate-controlling polymer to provide a homogenous or intimate mixture.

[72] In another embodiment, the application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition is in a monolithic matrix form and comprises intra-granular and extra-granular portions.

[73] In another embodiment, the application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition is in a monolithic matrix form comprising at least one hydrophilic polymer present in both extra-granular portion and intra-granular portions.

[74] In another embodiment, the application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable

salt thereof, wherein said composition is in a monolithic matrix form comprising at least one hydrophilic polymer present in at least in intra-granular or extra-granular portions.

[75] In certain embodiments, the application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein the said composition comprises at least one hydrophilic polymer present in both intra-granular and extra-granular portions in a weight ratio of about 0:1 to 1:1.

[76] In certain embodiments, the rate-controlling polymer comprises at least one hydrophilic polymer in an amount of from about 10% to about 70% by weight based on the total weight of the composition.

[77] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said rate-controlling polymer comprises at least one hydrophilic polymer in an amount of from about 10% to about 70% by weight based on the total weight of the composition.

[78] In yet another embodiment, the present application relates to a modified release composition for once daily oral administration comprising from about 30 mg to 60 mg of nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said rate-controlling polymer comprises at least one hydrophilic polymer in an amount of from about 10% to about 70% by weight based on the total weight of the composition and said composition provides therapeutically effective amount of nicergoline for a period of about 24 hours after oral administration.

[79] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically

acceptable salt thereof, wherein said composition releases no more than about 30% of nicergoline in 2 hours, in 900ml of 0.1 N HCl solution, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C.

[80] In certain embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition releases no more than about 30% of nicergoline in 2 hours, in 900ml of acetate buffer having pH of 4.5, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C.

[81] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition releases no more than about 30% of nicergoline in 2 hours, in 900ml of phosphate buffer having pH of 6.8, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C.

[82] In certain embodiment, the composition of the present application may be subjected to changeover dissolution study which is performed by using USP apparatus I (Basket) with 100 RPM, and the dissolution methodology includes change over dissolution mediums i.e., one acid stage followed by two buffer stages. Firstly in-vitro dissolution was carried out in acid stage (900 ml of 0.1N HCl solution, USP Type 1 apparatus at a speed of 100 rpm and 37°C) till 60 minutes, followed by first buffer stage (900 ml of acetate buffer solution having pH 4.5 at a speed of 50 rpm and 37°C) and second buffer stage (900 ml of phosphate buffer solution having pH 6.8 a speed of 100 rpm and 37°C).

[83] In certain embodiment, a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when subjected to a changeover dissolution study releases no more than

about 30% in 2 hours, no more than about 60% in 6 hours, and more than about 70% in 10 hours of nicergoline.

[84] In another embodiment, a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition exhibits at least one of the following dissolution profiles:

(i) releases no more than about 30% of nicergoline in 2 hours, in 900ml of 0.1 N HCl solution, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C,

(ii) releases no more than about 30% of nicergoline in 2 hours, in 900ml of acetate buffer having pH of 4.5, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C,

(iii) releases no more than about 30% of nicergoline in 2 hours, in 900ml of phosphate buffer having pH of 6.8, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C and

(iv) releases no more than about 30% in 2 hours, no more than about 60% in 6 hours, and more than about 70% in 10 hours of nicergoline when subjected to a changeover dissolution study.

[85] In one embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline 30 mg or its pharmaceutically acceptable salt thereof, wherein said composition exhibits bioequivalence to a commercially available 10 mg nicergoline immediate release composition when administered thrice daily to a human subject in fasting conditions.

[86] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when administered as

single dose exhibits bioequivalence to a commercially available 10 mg nicergoline immediate release composition when administered thrice daily to a human subject in fasting conditions and said bioequivalence is established by having 90% confidence interval (CI) of the relative mean $AUC_{(0-t)}$ within 80.00% to 125.00%.

[87] In yet another embodiment, the present application relates to a modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when administered as single dose exhibits bioequivalence to a commercially available 10 mg nicergoline immediate release composition when administered thrice daily to a human subject in fasting conditions and said bioequivalence is established by having 90% confidence interval (CI) of the relative mean $AUC_{(0-\infty)}$ within 80.00% to 125.00%.

[88] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when administered as a single dose to a human subject in the fasting and fed conditions have a mean AUC_{0-4} in the ratio of about 1:1.3.

[89] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when administered as a single dose to a human subject in the fasting conditions have an exposure up to 8, 16 and 24 hours similar to an immediate release formulation comprising nicergoline 10 mg administered thrice daily in fasting conditions and have the mean AUC_{0-8}, AUC_{0-16} and AUC_{0-24} in the ratio of about 1:1.

[90] In certain embodiments, the application relates to a modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its

pharmaceutically acceptable salt thereof, wherein said composition when administered as a single dose to a human subject in the fasting conditions have an average peak to trough plasma concentration ratio of simulated steady state profiles of about 1.66:1.

[91] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when administered as a single dose to a human subject in the fasting conditions have a mean AUC_{0-8} to AUC_{0-2} in the ratio of about 1: 0.03.

[92] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when administered as a single dose to a human subject in the fasting conditions and in the fed conditions have a plasma concentration profile at 2 hours in the ratio of about 1:1.

[93] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof and at least one rate-controlling polymer, wherein the said composition following single dose oral administration to a human subject in fasting conditions exhibit about 6% of nicergoline absorption profile at 2 hours.

[94] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof and at least one rate-controlling polymer, wherein said composition following single dose oral administration to a human subject in fasting conditions exhibit about 20% of nicergoline absorption profile at 4 hours.

[95] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically

acceptable salt thereof and at least one rate-controlling polymer, wherein said composition following single dose oral administration to healthy adult humans in fasting conditions exhibit about 42% of the nicergoline absorption profile at 8 hours.

[96] In another embodiment, the present application relates to a modified release composition for once daily oral administration to a human subject comprising nicergoline or its pharmaceutically acceptable salt thereof and at least one rate-controlling polymer, wherein said composition following single dose oral administration to healthy adult humans in fasting conditions exhibit about 58% of the nicergoline absorption profile at 12 hours.

[97] In another embodiment, the present application relates to a modified release composition for once daily oral administration to a human subject comprising nicergoline or its pharmaceutically acceptable salt thereof and at least one rate-controlling polymer, wherein said composition following single dose oral administration to healthy adult humans in fasting conditions exhibits at least one of the following plasma profiles:

(i) about 6% of nicergoline absorption profile at 2 hours,

(ii) about 20% of nicergoline absorption profile at 4 hours,

(iii) about 42% of the nicergoline absorption profile at 8 hours, and

(iv) about 58% of the nicergoline absorption profile at 12 hours.

[98] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline 30 mg or its pharmaceutically acceptable salt thereof, wherein the composition exhibits a simulated mean C_{24}/C_{max} ratio of MDL of about 0.6:1.0 when administered to healthy volunteers at steady-state (for 7 days).

[99] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline 30 mg or its pharmaceutically acceptable salt thereof, wherein the composition exhibits a simulated steady

state AUC, which is in the range from 80% to 125% of the AUC of the same amount of nicergoline administered as an immediate release formulation thrice daily.

[100] In another embodiment, the present application relates to a modified release composition for once daily oral administration to a human subject comprising nicergoline 30 mg, wherein said composition has delayed T_{max} in both fasting and fed conditions when compared with first dose of the three divided doses of commercially available 10 mg nicergoline immediate release composition.

[101] In another embodiment, the present application relates to a modified release composition for once daily oral administration to a human subject comprising nicergoline 30 mg, wherein the said composition has T_{max} in the range of about 5-20 hours.

[102] In further embodiment, the composition has T_{max} of about 8-15 hours or about 10-15 hours or about 13 hours.

[103] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said dosage form has C_{max} of about 5-19 ng/ml of nicergoline or its detectable metabolites when administered to human subjects in fasting condition.

[104] In further embodiment, the composition has C_{max} of about 7-15 ng/ml or about 8-15 ng/ml or about 8-10 ng/ml of nicergoline or its detectable metabolites.

[105] In another embodiment, the present application relates to a modified release composition for once daily oral administration to a human subject comprising about 30 mg of nicergoline, wherein said composition when administered to a human subject under fasting conditions provides steady state $C_{min,ss}$ of about 8 ng/ml to about 15 ng/ml of nicergoline or its detectable metabolites.

[106] In another embodiment, the present application relates to a modified release composition for once daily oral administration to a human subject comprising about 30 mg of

nicergoline, wherein said composition when administered to a human subject under fasting conditions provides steady state $C_{max,ss}$ of about 13 ng/ml to about 26 ng/ml of nicergoline or its detectable metabolites.

[107] In another embodiment, the present application relates to a modified release composition for once daily oral administration to a human subject comprising about 30 mg of nicergoline, wherein said composition when administered to a human subject under fasting conditions provides steady state $C_{avg,ss}$ of about 10 ng/ml to about 18 ng/ml of nicergoline or its detectable metabolites.

[108] In yet another embodiment, the present application relates to a modified release composition for once daily oral administration to a human subject comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when administered to a human subject under fasting conditions provides steady state plasma profile and exhibits at least one of the following plasma profiles:

(i) C_{max} of about 5 ng/ml to 19 ng/ml of nicergoline or its detectable metabolites,

(ii) $C_{min,ss}$ of about 8 ng/ml to about 15 ng/ml of nicergoline or its detectable metabolites,

(iii) $C_{max,ss}$ of about 13 ng/ml to about 26 ng/ml of nicergoline or its detectable metabolites, and

(iv) $C_{avg,ss}$ of about 10 ng/ml to about 18 ng/ml of nicergoline or its detectable metabolites.

[109] In certain embodiments of the present application, the rate-controlling polymer used may be hydrophilic, hydrophobic or combinations thereof. The rate-controlling polymers are meant to control/ sustain the release rate of nicergoline.

[110] Suitable examples of hydrophilic polymers which can be used in the present application, include, but are not limited to, cellulose derivatives such as hydroxypropylcellulose, hydroxypropylmethylcellulose, hydroxyethylcellulose,

hydroxymethylcellulose, carboxymethylcellulose, methylcellulose, sodium carboxy methylcellulose or combinations thereof; non-cellulose polysaccharides (galactomannans, guar gum, carob gum, gum arabic, sterculia gum, agar, pregelatinized starch, alginates and the like); polyvinylpyrrolidone; polyvinyl acetate; copolymer of vinylpyrrolidone and vinyl acetate, polyalkylene glycols, acrylic acid polymers such as cross linked acrylic acid-based polymers; and a mixture comprising one or more of said materials.

[111] In certain embodiments, hydroxypropylmethylcellulose polymers include those sold under the Trademark of METHOCEL® (DoW Chemical Corporation) or under the Trademark of METOLOSE® (Shin-Etsu). The polymers can be of different grades and having varying viscosities.

[112] Suitable examples of hydrophobic polymers which can be used in the present application, include, but are not limited to, bees wax, glycol wax, carnauba wax, hydrogenated vegetable oil, hydrogenated castor oil, hydrogenated cottonseed oil, hydrogenated palm oil, hydrogenated soyabean oil, cetyl alcohol, cetostearyl alcohol, lauryl alcohol, palmityl alcohol, tribehenin, fatty acids, fatty acid triglycerides, stearic acid, ethyl cellulose, glycerol palmitostearate, glycerol monostearate, stearylalcohol, glyceryl behenate, polyanhydrides, polyacrylates, polymethacrylates polyvinyl acetate, eudragit, cellulose derivatives such as ethyl cellulose, propyl cellulose cellulose acetate, cellulose propionate, cellulose acetate-butyrate, cellulose acetate propionate or combinations thereof.

[113] In another embodiment of the present application, the modified release composition for once daily oral administration of nicergoline, further comprises one or more other pharmaceutically acceptable excipients such as one or more of diluents, binders, glidants, lubricants, colouring agents and coating materials.

[114] Suitable examples of diluents which can be used in the present application include, but are not limited to, starches, lactose, mannitol, Pearlitol™ SD 200, celluloses,

confectioner's sugar, and the like. Different grades of lactose can be used and it includes, but are not limited to, lactose monohydrate, lactose DT (direct tabletting), lactose anhydrous, Flowlac™ (available from Meggle Products), Pharmatose™ (available from DMV), and others. Different grades of starch can be used and it includes, but are not limited to, maize starch, potato starch, rice starch, wheat starch, pregelatinized starches (commercially available as PCS PC10 from Signet Chemical Corporation) and Starch 1500, Starch 1500 LM grade (low moisture content grade) from Colorcon, fully pregelatinized starches (commercially available as National 78-1551 from Essex Grain Products) and others. Different grades of celluloses can be used and it includes crystalline celluloses or powdered celluloses. Examples of crystalline cellulose products include, but are not limited to, CEOLUS™ KG801, Avicel™ PH101, PH102, PH301, PH302 and PH-F20, microcrystalline cellulose 114, and microcrystalline cellulose 112. Other useful diluents include, but are not limited to, carmellose, sugar alcohols such as mannitol, sorbitol and xylitol, calcium carbonate, magnesium carbonate, dibasic calcium phosphate, directly compressible grades of dibasic calcium phosphate (Emcompress™) and tribasic calcium phosphate. Useful binders include, but are not limited to, hydroxypropyl celluloses (Klucel™ LF), Nisso HPC-L (Nippon Soda Co. Ltd.), HPC-EXF, low viscosity hydroxypropyl methylcelluloses (HPMC or hypromelloses, e.g., Methocel™), polyvinylpyrrolidones or povidones (e.g., grades PVP-K25, PVP-K29, PVP-K30, PVP-K90, etc.), copovidones, (e.g., Plasdone™ S 630, a random copolymer of N- vinyl-2-pyrrolidone and vinyl acetate), powdered acacia, gelatin, guar gum, carbomers (e.g. Carbopol™ products), water-soluble methylcelluloses, polymethacrylates, and starches.

[115] Suitable examples of binders which can be used in the present application include, but are not limited to, carboxymethylcelluloses, hydroxyethylcelluloses, dextrin, gelatin, maltodextrin, polyethylene oxides, sodium alginate, hydroxypropylcelluloses, hydroxypropyl

methylcelluloses, polyvinylpyrrolidones or povidone (e.g., PVP-K25, PVP-K29, PVP-K30, and PVP-K90D), powdered acacia, gelatin, guar gum, carbomers (e.g., a Carbopol™ product), methylcelluloses, polymethacrylates, and starches.

[116] Various glidants or anti-sticking agents can be used in the present application include, but are not limited to, talc, silica derivatives, colloidal silicon dioxide and the like or mixtures thereof.

[117] Various lubricants that can be used in the present application include, but are not limited to, stearic acid, magnesium stearate, calcium stearate, sodium lauryl sulphate, hydrogenated vegetable oil, hydrogenated castor oil, sodium stearyl fumarate, talc, glyceryl behenate, glyceryl monostearates, palmitic acid, carnauba wax, calcium soaps, zinc stearate, polyoxyethylene monostearates, calcium silicate, silicon dioxide, macrogols, and mixtures thereof.

[118] Various useful colorants that can be used in the present application include any FDA approved colours including, but are not limited to, Food Yellow No. 5, Food Red No. 2, Food Blue No. 2, food lake colorants, iron oxides and the like or mixtures thereof.

[119] The modified release compositions of the present application may be prepared by using any conventional techniques, not limiting to, dry granulation, wet granulation, direct compression, or combinations thereof.

[120] In some embodiments, the application relates to a modified release composition comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition is prepared by processes known in the art including, direct compression, dry granulation, wet granulation, fluidized bed granulation or combinations of such processes.

[121] In another embodiment, the present application relates to a process of preparing modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, comprising the steps of:

a) blending the desired quantity of nicergoline with intra granular excipients to form uniform blend,

b) granulating the blend of step (a) with binder solution to form nicergoline granules,

c) mixing the nicergoline granules of step (b) with extra granular excipients,

d) compressing the mixture of step (c) into tablets, and

e) optionally film coating the tablets of step (d) to attain desired weight.

[122] In certain embodiments, the present application may employ various granulation methods, for example, a) wet granulation methods, in which a binder and a solvent are added to a powder mixture and granulation is conducted, b) dry granulation methods, in which granulation is carried out by compacting/ decompacting or slugging/ deslugging a powder mixture, and c) melt granulation methods, in which a substance that melts on heating is mixed with an active agent and granulation is carried out.

[123] There are no particular limitations on employing granulating solvents in this application. The solvent may be water or any of various organic solvents, for example, lower alcohols such as methanol and ethanol, ketones such as acetone and methyl ethyl ketone, methylene chloride, or any mixtures thereof.

[124] For preparing granules operations such as mixing granulation methods employ planetary mixer, a screw mixer and the like; high-speed mixing granulation methods using a Henschel mixer, a Super mixer and the like; extruding granulation methods using a cylindrical granulator, a rotary granulator, a screw extruding granulator, a pellet mill type granulator and the like; wet high-shear granulation methods, fluidized-bed granulation methods, compression granulation methods, crushing granulation methods and spraying granulation methods can be used.

[125] After granulation, drying operation can be performed using an oven dryer, a fluidized bed dryer and the like; post drying crushing and sieving can be carried out to obtain granules or fine granules for use.

[126] Equipment suitable for processing pharmaceutical compositions of the present application include any one or a combination of mechanical sifters, blenders, roller compactors, granulators (rapid mixer or fluid bed granulator), fluid bed dryers, compression machines, rotating bowls or coating pans, etc.

[127] The composition of present application may include one or more film coatings over the final composition.

[128] The composition of present application may be prepared as tablets, capsules, sachets, pills, or granules.

[129] In certain embodiments, the application relates to a modified release composition for once daily oral administration comprising from about 30 mg to 60 mg of nicergoline or its pharmaceutically acceptable salt thereof nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition is packaged in a strip or a blister or a HDPE container optionally together with a desiccant and/or oxygen absorbent.

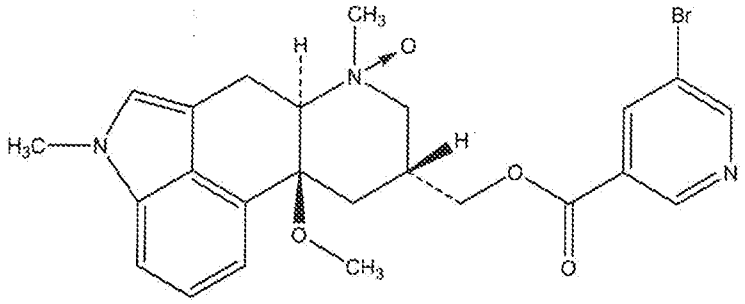
[130] Suitable examples of packaging materials used in the present application include, but are not limited to, containers including lids, made of polyethylene and/or polypropylene and/or glass, and blisters or strips made of aluminium or high-density polypropylene, or polyvinyl chloride, or polyvinyl chloride coated with polyvinylidene dichloride, generally termed PVC/PVDC. The packaging material optionally utilizes oxygen absorbers or desiccants with packaging material.

[131] In another embodiment, the present application provides a modified release composition comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein

said composition remains physically and chemically stable, upon storage at a temperature of about 40°C and relative humidity of about 75% for at least three months.

[132] In yet an another embodiment, the application provides a modified release composition comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition upon storage at a temperature of about 40°C and relative humidity of about 75% for three months produces not more than 3% of total relative substances.

[133] In another embodiment, the present application relates to a modified release composition comprising about 30 mg to 60 mg of nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said composition upon storage at a temperature of about 40°C and relative humidity of about 75% for three months produces less than 0.5% w/w of nicergoline N-Oxide compound having the structure of:



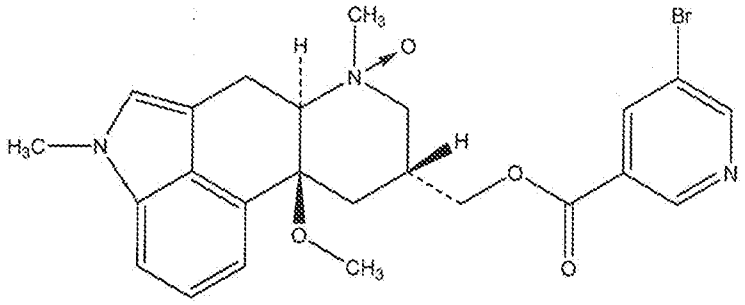
[134] In yet another embodiment, the present application relates to a process for producing a validated batch of a modified release composition containing nicergoline or its pharmaceutically acceptable salt thereof, comprising:

a) producing a batch of modified release composition comprising about 30 mg to 60 mg of nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients,

b) performing stability testing with a sample of composition from the produced batch,

c) determining the total amount of nicergoline N-Oxide compound present in the batch; and

d) validating the batch for distribution, only if the sample of the batch after stability testing contains less than 0.5% w/w of nicergoline N-Oxide compound, wherein the nicergoline N-Oxide compound have the structure of:



[135] In a certain aspect of the embodiments, validation of the composition of present application after stability testing is usually determined by a RP-HPLC method.

[136] In another embodiment, the present application relates to a modified release composition for once daily oral administration comprising nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition is administered for the treatment of patients having acute or chronic cerebral metabolic-vascular disorders having cognitive loss and/ or dementia.

[137] In yet another embodiment, the present application relates to a method for treating patients having acute or chronic cerebral metabolic-vascular disorders having cognitive loss and/or dementia, comprising orally administering an once daily modified release composition comprising from about 30 mg to 60 mg of nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients, wherein said composition provides therapeutically effective amount of nicergoline for a period of about 24 hours after oral administration.

[138] The present application is further illustrated by the examples which are provided merely to be exemplary of the composition described above and do not limit the scope of the application. Certain modifications and equivalents will be apparent to those skilled in the art and are intended to be included within the scope of the present application.

[139] **EXAMPLES**

[140] **Table-1**

SL.	Ingredients	Example-1	Example-2
		%	%
Intragranular portion			
1	Nicergoline	7.50	6.98
2	Lactose Monohydrate	20.50	8.14
3	Microcrystalline Cellulose	20.50	8.14
4	Polyvinylpyrrolidones (PVPK-30)	8.00	8.37
5	Hydroxypropylmethyl cellulose (Methocel K100 LVCR)		27.90
6	Purified Water	q.s.	q.s.
Extragranular portion			
7	Hydroxypropylmethyl cellulose (Methocel K100 LVCR)	15.00	30.23
8	Hydroxypropylmethyl cellulose (Methocel K4 MCR)	6.00	
9	Microcrystalline Cellulose	21.50	
10	Lactose Monohydrate		9.30
11	Magnesium stearate	1.00	0.93
	Total	100%	100%
12	Opadry	@ 3%	@ 3%

[141] **Manufacturing Process:**

a) Nicergoline and all the intra granular excipients are shifted together through a suitable sieve and further transferred to a suitable mixer granulator

b) The mixture of step (a) is granulated using binder solution

c) The granules of step (b) were dried in a suitable drier and milled further to suitable size

d) The mixture of step (c) is further mixed with extra granular excipients and compressed to tablets using suitable tooling

e) The tablets of step (d) are further coating with opadry using suitable solvents.

[142] **Example- 3:**

[143] **Dissolution Study:**

[144] The dissolution of the modified release composition containing nicergoline was performed in different change over medias i.e. 1.0 hour in 0.1N HCl, 3.0 hours in pH 4.5 acetate buffer followed by pH 6.8 phosphate buffer up to 16 hours and the dissolution was carried out in USP type I apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C. The results are shown in Fig 1.

[145] **Example- 4:**

[146] **Pharmacokinetic study:**

[147] An open label, four-way crossover pharmacokinetic study was conducted involving administration of the nicergoline composition of Example-1 and Example- 2 containing 30 mg nicergoline as test products, given once daily and SERMION® oral tablets (containing nicergoline 10 mg as immediate release) from Pfizer, U.K. (reference product) given thrice daily (t.i.d.) under fasting condition for first three periods with 17 healthy human volunteers and plasma concentrations of the drug compounds were determined at desired intervals after dosing. Pharmacokinetic data obtained in the study is shown in table 2 & 3, and Fig. 2-4.

[148] **Table-2: Pharmacokinetic data for nicergoline – Example-1**

Pharmacokinetic data	AUC_{0-t}	$AUC_{0-\infty}$	C_{max}	T_{max} after single dose administration
Sermion® 10 mg Tablets (Reference product, R)	29.12 ng.hr/ml	30.51 ng.hr/ml	10.90 ng.hr/ml	4 Hours
Example-1 tablets (Test product, T1)	26.54 ng.hr/ml	28.55 ng.hr/ml	9.88 ng.hr/ml	8 Hours
Ratio of geometric least square means (T/R) %	91	94	91	-
90% confidence intervals (T/R) %				
Lower limit	83	86	77	-
Upper limit	100	102	106	-

[149] **Table-3: Pharmacokinetic data for nicergoline – Example-2**

Pharmacokinetic data	AUC_{0-t}	$AUC_{0-\infty}$	C_{max}	T_{max} after single dose administration
Sermion® 10 mg Tablets (Reference product, R)	29.12 ng.hr/ml	30.51 ng.hr/ml	10.90 ng.hr/ml	4 Hours
Example-2 tablets (Test product, T2)	26.26 ng.hr/ml	28.2 ng.hr/ml	8.95 ng.hr/ml	13 Hours
Ratio of Least square means (T/R) %	90	92	82	-
90% confidence intervals (T/R) %				
Lower limit	82	85	71	-
Upper limit	99	101	96	-

[150] **Example- 5:**

[151] **Stability Study:**

[152] The pharmaceutical compositions of the present application Example-1 and Example-2 were subjected to stability study at 40°C/75% RH for six months. The initial and stored samples were analyzed for impurities of nicergoline. The impurity data obtained in the study is shown in the following tables- 4 and 5

[153] **Table-4- Stability data of Example-1 in PVC/PVDC Blister Packs.**

Period (months)	Conditions	Impurity Details (% RS)										
		(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)	Single Highest Unknown	Total
Initial	25°C	0.16	0.03	0.08	0.05	ND	0.11	0.04	0.08	0.17	0.06	0.87
1M	40°C/75%RH	0.14	0.03	0.18	0.08	ND	0.10	0.03	0.11	0.32	0.15	1.16
	30°C/65%RH	0.14	0.02	0.10	0.04	ND	0.10	0.03	0.06	0.21	0.08	0.82
	25°C/60%RH	0.16	0.02	0.08	0.03	ND	0.11	0.03	0.08	0.18	0.04	0.77
2M	40°C/75%RH	0.12	0.04	0.33	0.15	ND	0.07	0.04	0.09	0.34	0.18	1.40
3M	40°C/75%RH	0.14	0.03	0.74	0.57	ND	0.07	0.02	0.14	0.30	0.03	2.22
	30°C/65%RH	0.14	0.03	0.18	0.14	ND	0.09	0.02	0.10	0.24	0.09	1.06
	25°C/60%RH	0.14	0.03	0.10	0.07	ND	0.10	0.02	0.09	0.19	0.07	0.89
6M	40°C/75%RH	0.15	0.03	2.25	1.59	0.01	0.08	0.03	0.12	0.15	0.07	4.66
	30°C/65%RH	0.16	0.04	0.26	0.18	0.00	0.11	0.03	0.10	0.33	0.13	1.51
	25°C/60%RH	0.16	0.04	0.15	0.09	0.00	0.11	0.03	0.09	0.27	0.09	1.19

ND- Not Detected

Impurity (A) - Chloro-nicergoline

Impurity (B) -Nor-nicergoline

Impurity (C) -MeLUOL

Impurity (D) -Bromo-nicotinic Acid

Impurity (E) -10 Alpha-hydroxy nicergoline

Impurity (F) -α-α-Nicergoline

Impurity (G) - Dihydro-nicergoline

Impurity (H) - Desmethyl-nicergoline

Impurity (I) - Nicergoline-N-Oxide

[154] **Table-5- Stability data -Example-2 in PVC/PVDC Blister Packs.**

Period (months)	Conditions	Impurity Details (% RS)										
		(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)	Single Highest Unknown	Total
Initial	25°C	0.16	0.02	0.07	0.02	ND	0.10	0.04	0.05	0.07	0.04	0.63
1M	40°C/75%RH	0.16	0.02	0.10	0.03	ND	0.10	0.05	0.05	0.12	0.04	0.72
	30°C/65%RH	0.17	0.02	0.07	0.02	ND	0.08	0.06	0.05	0.09	0.07	0.66
	25°C/60%RH	0.16	0.03	0.06	0.02	ND	0.09	0.03	0.05	0.08	0.03	0.56
2M	40°C/75%RH	0.17	0.02	0.12	0.05	ND	0.10	0.05	0.06	0.20	0.08	0.86
3M	40°C/75%RH	0.14	0.02	0.24	0.16	ND	0.10	0.02	0.08	0.21	0.08	1.12
	30°C/65%RH	0.15	0.02	0.09	0.04	ND	0.10	0.02	0.06	0.09	0.04	0.68
	25°C/60%RH	0.14	0.03	0.07	0.03	ND	0.10	0.01	0.06	0.07	0.03	0.61
6M	40°C/75%RH	0.16	0.02	0.79	0.54	0.01	0.10	0.03	0.10	0.15	0.04	2.07
	30°C/65%RH	0.16	0.03	0.12	0.05	0.00	0.11	0.03	0.06	0.15	0.06	0.93
	25°C/60%RH	0.16	0.03	0.06	0.03	0.01	0.11	0.03	0.05	0.11	0.04	0.77

ND- Not Detected

Impurity (A) - Chloro-nicergoline

Impurity (B) -Nor-nicergoline

Impurity (C) -MeLUOL

Impurity (D) -Bromo-nicotinic Acid

Impurity (E) -10 Alpha-hydroxy nicergoline

Impurity (F) - α-α-Nicergoline

Impurity (G) - Dihydro-nicergoline

Impurity (H)- Desmethyl-nicergoline

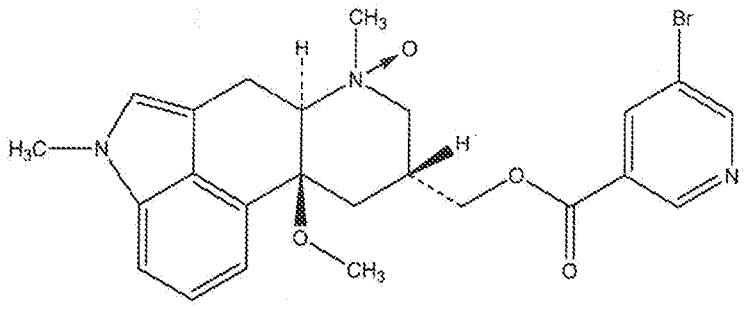
Impurity (I) - Nicergoline-N-Oxide

[155] The compounds stated as impurities in European Pharmacopeia, Monograph of Nicergoline are analyzed by RP-HPLC method. Relative retention times for these compounds shown in the following table- 6

[156] **Table-6 -** Relative retention times of nicergoline and its impurities.

Name	**RRT(Approx)**
Impurity-C	0.18
Impurity-B	0.68
Impurity-E	0.88
Impurity-H	0.90
Impurity-A	0.97
Nicergoline	1.0
Impurity-G	1.09
Impurity-F	1.14

[157] Nevertheless, one more impurity called Nicergoline N-oxide, which is not present in pharmacopeia was identified. This compound is chemically (6aR,9R,10aS) -9-(((5-bromonicotinoyl)oxy)methyl)-10a-methoxy-4,7-dimethyl-4,6,6a,7,8,9,10,10a-octahydroindolo[4,3-fg]quinolone 7-oxide compound and having the structure of



[158] Relative retention time of this compound was found to be 0.35 in RP-HPLC

[159] **Table-7 -** Relative retention time of Nicergoline N-oxide impurity

Name	**RRT(Approx.)**
Nicergoline N-oxide	0.35

[160] The nicergoline and its impurities were estimated by RP-HPLC method, by using pH 7.5 phosphate buffer and two mobile phases A & B were prepared by using acetonitrile, water and pH 7.5 phosphate buffer.

[161] Although the invention has been illustrated by certain of the preceding examples, it is not to be construed as being limited thereby; but rather, the invention encompasses the generic area as hereinbefore disclosed. Various modifications and embodiments can be made without departing from the spirit and scope thereof.

Claims:

1. A modified release composition for once daily oral administration comprising from about 30 mg to 60 mg of nicergoline or its pharmaceutically acceptable salt thereof, at least one rate-controlling polymer and one or more other pharmaceutically acceptable excipients,

   wherein said rate-controlling polymer comprises at least one hydrophilic polymer in an amount of from about 10% to about 70% by weight based on the total weight of the composition.

2. The composition of claim 1, wherein said composition is in a monolithic matrix form and comprises intra-granular and extra-granular portions.

3. The composition of claim 1, wherein said hydrophilic polymer is present in intra-granular and extra-granular portions in a weight ratio of about 0:1 to 1:1.

4. The composition of claim 1, wherein said composition exhibits at least one of the following dissolution profiles:

   (i) releases no more than about 30% of nicergoline in 2 hours, in 900ml of 0.1 N HCl solution, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C,

   (ii) releases no more than about 30% of nicergoline in 2 hours, in 900ml of acetate buffer having pH of 4.5, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C,

   (iii) releases no more than about 30% of nicergoline in 2 hours, in 900ml of phosphate buffer having pH of 6.8, when tested in USP type 1 apparatus at a speed of 100 rpm and at a temperature of 37°C ± 0.5°C and

(iv) releases no more than about 30% in 2 hours, no more than about 60% in 6 hours, and more than about 70% in 10 hours of nicergoline when subjected to a changeover dissolution study.

5. The composition of claim 1, wherein said composition provides therapeutically effective amount of nicergoline for a period of about 24 hours after oral administration.

6. A modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition when administered as single dose exhibits bioequivalence to a commercially available 10 mg nicergoline immediate release composition when administered thrice daily to a human subject in fasting conditions and said bioequivalence is established by:

   a) a 90% Confidence Interval for mean $AUC_{(0-t)}$, which is between 80% and 125%, and

   b) a 90% Confidence Interval for mean $AUC_{(0-\infty)}$, which is between 80% and 125%.

7. The composition of claim 6, wherein said composition when administered as a single dose in the fasting and fed conditions have a mean AUC_{0-4} in the ratio of about 1:1.3.

8. The composition of claim 6, wherein said composition when administered as a single dose in the fasting and fed conditions have the mean AUC_{0-8}, AUC_{0-16} and AUC_{0-24} in the ratio of about 1:1.

9. The composition of claim 6, wherein said composition when administered as a single dose in the fasting conditions have an exposure up to 8, 16 and 24 hours similar to an immediate release formulation comprising nicergoline 10 mg administered thrice daily in fasting conditions.

10. The composition of claim 6, wherein said composition when administered as a single dose in the fasting conditions have an average peak to trough plasma concentration ratio of simulated steady state plasma profiles of about 1.66:1.
11. The composition of claim 6, wherein said composition when administered as a single dose in the fasting conditions have a mean AUC_{0-8} to AUC_{0-2} in the ratio of about 1: 0.03.
12. The composition of claim 6, wherein said composition when administered as a single dose in the fasting and fed conditions have a plasma concentration profile at 2 hours in the ratio of about 1:1.
13. The composition of claim 6, wherein said composition when administered as a single dose in the fasting conditions exhibit at least one of the following plasma profiles:
    (i) about 6% of nicergoline absorption profile at 2 hours,
    (ii) about 20% of nicergoline absorption profile at 4 hours,
    (iii) about 42% of the nicergoline absorption profile at 8 hours and
    (iv) about 58% of the nicergoline absorption profile at 12 hours.
14. A modified release composition for once daily oral administration comprising about 30 mg of nicergoline or its pharmaceutically acceptable salt thereof, wherein said composition has delayed T_{max} in both fasting and fed conditions when compared with first dose of the three divided doses of commercially available 10 mg nicergoline immediate release composition.
15. The composition of claim 14, wherein said composition has T_{max} of about 8-15 hours.
16. The composition of claim 14, wherein said composition when administered to a human subject under fasting conditions provides steady state plasma profile and exhibits at least one of the following plasma profiles:
    (i) C_{max} of about 5 ng/ml to 19 ng/ml of nicergoline or its detectable metabolites,

(ii) Cmin,ss is about 8 ng/ml to about 15 ng/ml of nicergoline or its detectable metabolites,

(iii) Cmax,ss is about 13 ng/ml to about 26 ng/ml of nicergoline or its detectable metabolites, and

(iv) Cavg,ss is about 10 ng/ml to about 18 ng/ml of nicergoline or its detectable metabolites.

17. The composition of claim 14, wherein said composition is administered for the treatment of patients having acute or chronic cerebral metabolic-vascular disorders.

18. The composition of claim 1, wherein said composition remains physically and chemically stable, upon storage at a temperature of about 40ºC and relative humidity of about 75% for at least three months.

19. The composition of claim 18, wherein said composition upon storage at a temperature of about 40ºC and relative humidity of about 75% for three months produces not more than 3% of total relative substances.

20. The composition of claim 18, wherein said composition upon storage at a temperature of about 40ºC and relative humidity of about 75% for three months produces less than 0.5% w/w of nicergoline N-Oxide compound having the structure of:

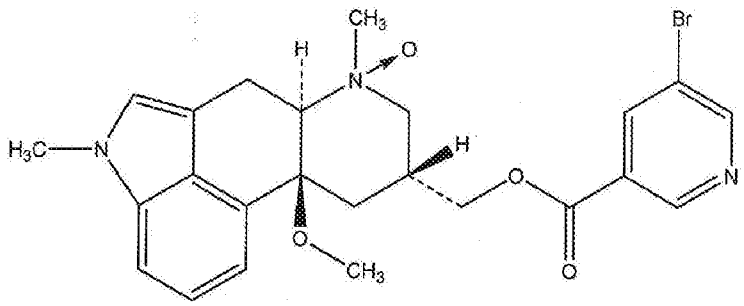


Figure.1

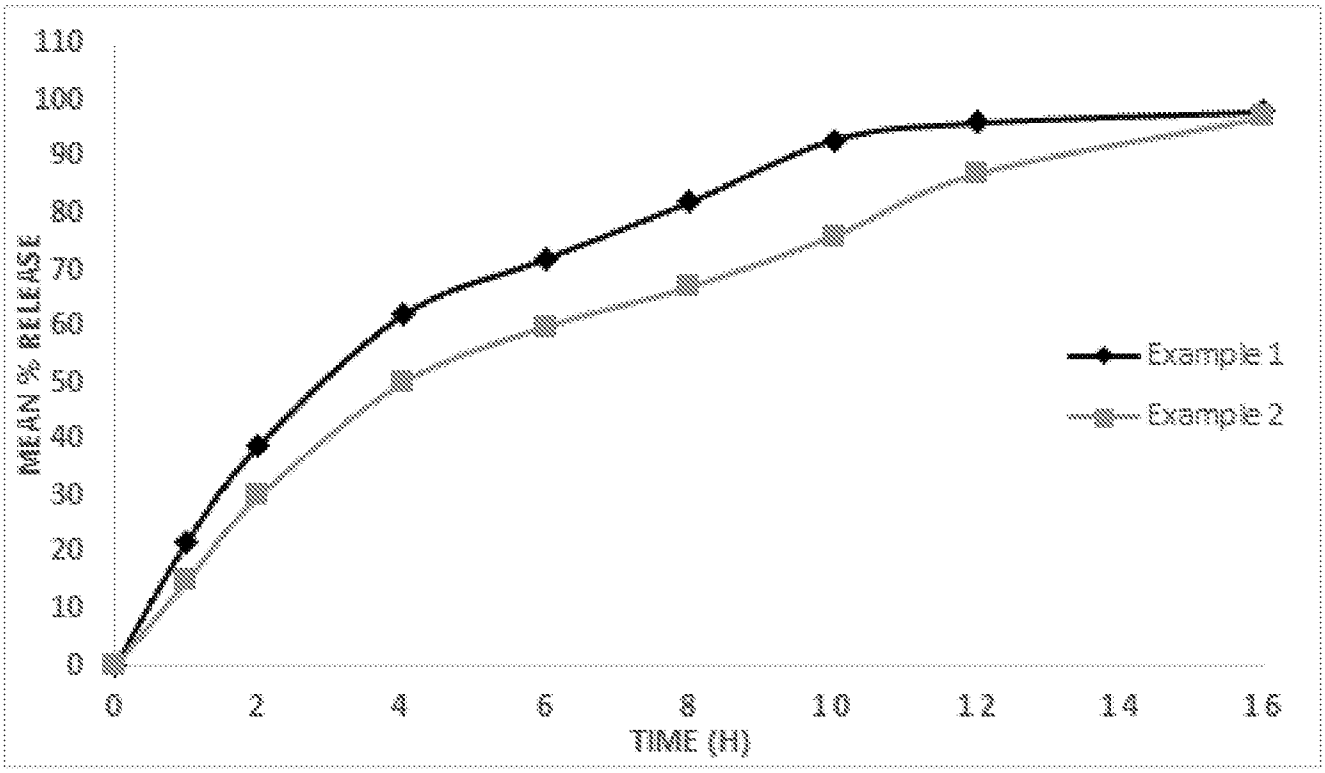

110
100
90
80
70
60
50
40
30
20
10
0
MEAN % RELEASE
0
2
4
6
8
10
12
14
16
TIME (H)
Example 1
Example 2

Figure-2

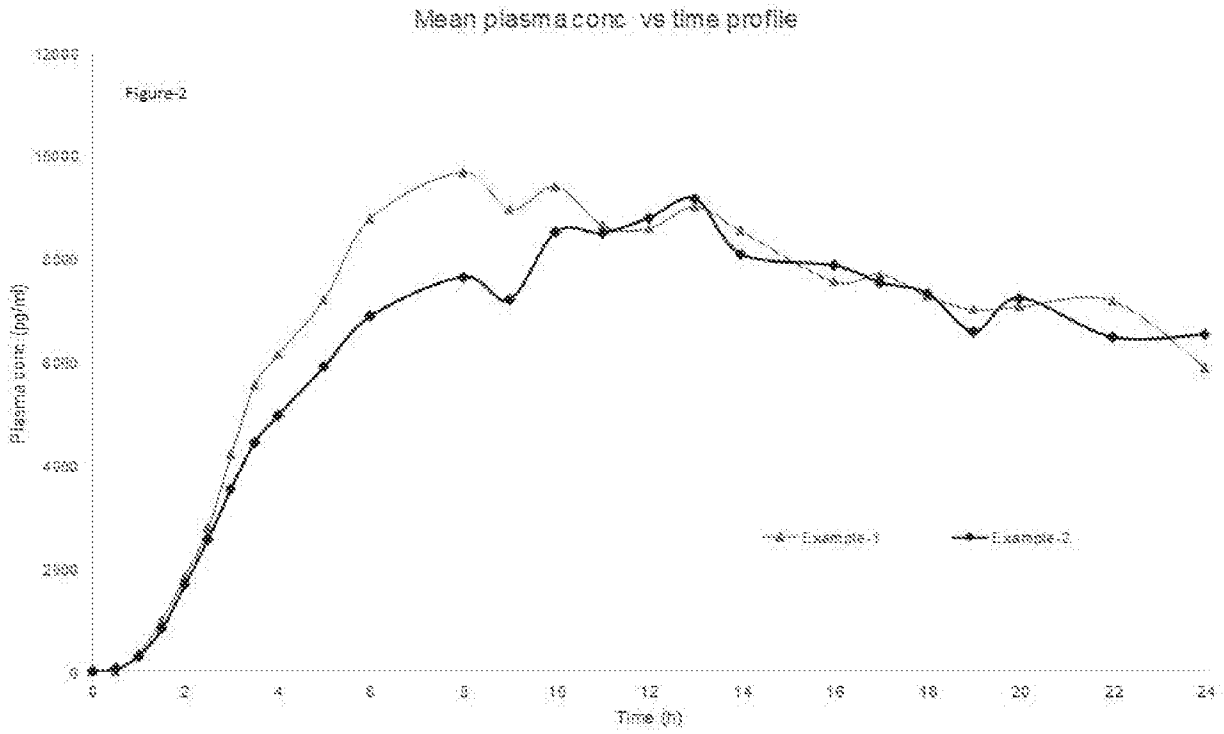

Mean plasma conc. vs time profile
Figure-2
Plasma conc. (pg/ml)
Time (h)
Example-1
Example-2

Figure-3

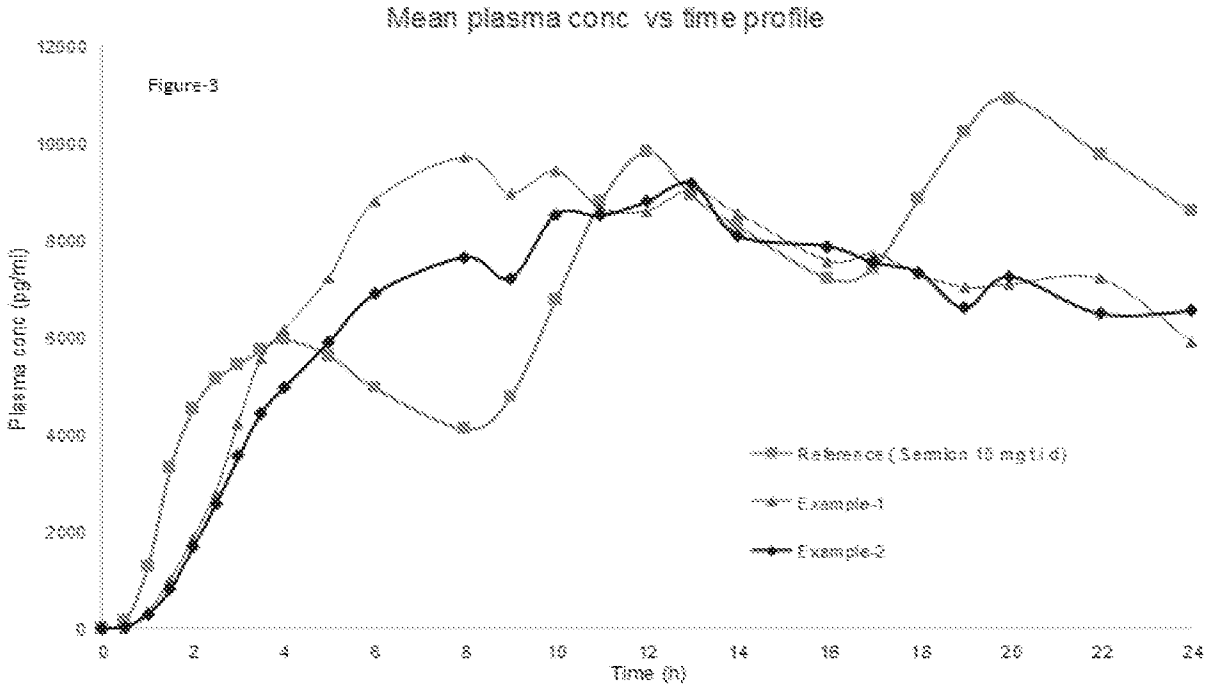
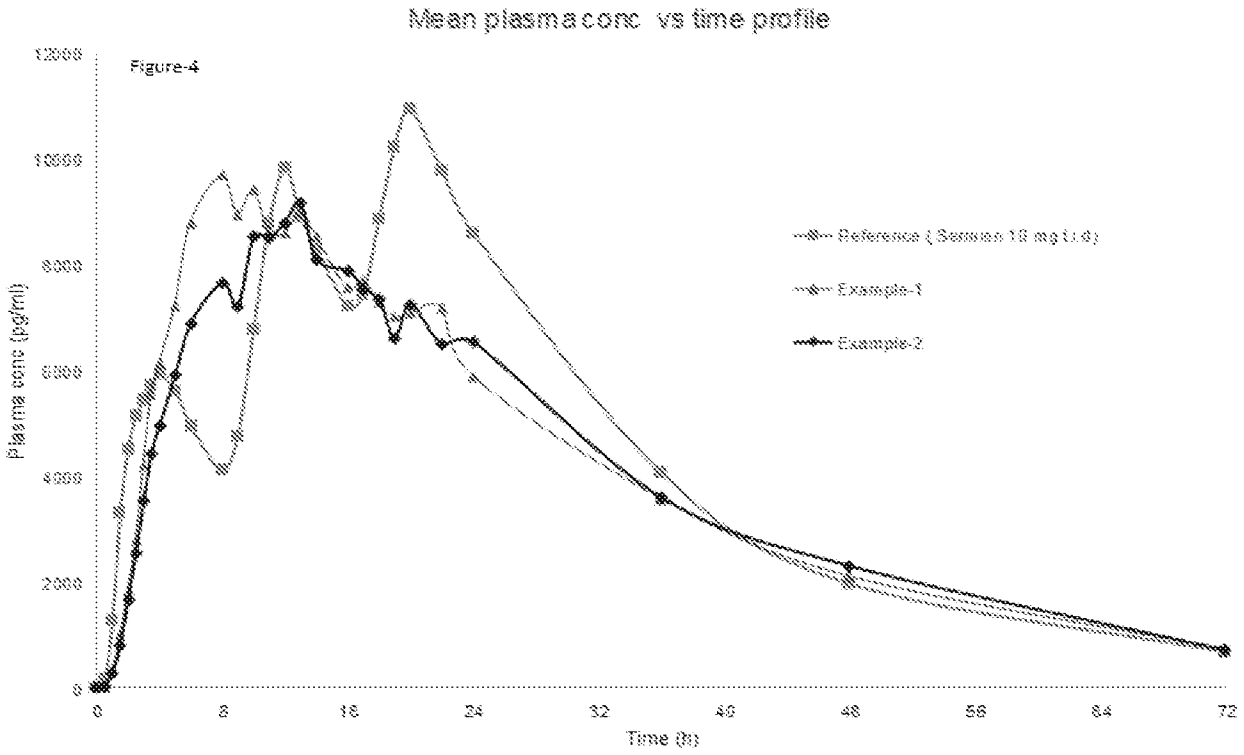

Mean plasma conc vs time profile
Figure-3
Plasma conc (pg/ml)
12000
10000
8000
6000
4000
2000
0
0
2
4
6
8
10
12
14
16
18
20
22
24
Time (h)
Example-1
Example-2

Figure-4


Mean plasma conc vs time profile
Figure-4
Plasma conc (pg/ml)
12000
10000
8000
6000
4000
2000
0
0
8
16
24
32
40
48
56
64
72
Time (h)
Reference (Semiron 18 mg i.v.)
Example-1
Example-2

INTERNATIONAL SEARCH REPORT

International application No
PCT/IB2016/054111

A. CLASSIFICATION OF SUBJECT MATTER
INV. A61K9/20 A61K31/48 A61P25/28
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
A61K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, BIOSIS, EMBASE, FSTA, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	RU 2 491 070 C2 (OOO FARMAMED [RU]; OOO NPF MIKROKHIM [UA]) 27 August 2013 (2013-08-27) page 5, lines 17-23 page 6, lines 27-31 examples 9, 13 -----	1-20
X	CN 102 949 372 A (SHANDONG FANGMING PHARM GROUP) 6 March 2013 (2013-03-06) Embodiment 3 claims -----	1-3,6-17

☐ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

* Special categories of cited documents :

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"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
17 October 2016	26/10/2016
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Authorized officer van de Wetering, P

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/IB2016/054111

Patent document cited in search report		Publication date	Patent family member(s)	Publication date
RU 2491070	C2	27-08-2013	NONE	
CN 102949372	A	06-03-2013	NONE	